Study 2: Group-level perceptions and metaperceptions

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Configuration

```
data_path <- "/Users/yaldasmac/Desktop/Metaperception/Study\ 2/study2_data.xlsx"

id_col       <- "ID"
ideology_col <- "ideology"
group_col       <- "group"
group_codes     <- c(`Pro-life` = 1, `Pro-choice` = 2)
group_text      <- c(`Pro-life` = "prolife", `Pro-choice` = "prochoice")
condition_col   <- "condition"
condition_codes <- c(Outgroup = 1, Metaperception = 2, Ingroup = 3)
condition_text  <- c(Outgroup = "outgroup", Metaperception = "metaperception",
                     Ingroup = "ingroup")

# MFQ-2 families
binding_subscales <- list(
  loyalty         = c("loyalty_1", "loyalty_2"),
  authority       = c("authority_1", "authority_2"),
  purity          = c("purity_1", "purity_2"),
  proportionality = c("proportionality_1", "proportionality_2")
)
individualizing_subscales <- list(
  care     = c("care_1", "care_2"),
  equality = c("equality_1", "equality_2")
)
```

```r
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(car)
library(emmeans)
library(effectsize)
library(knitr)
has_heplots <- requireNamespace("heplots", quietly = TRUE)
```

```r
fmt_p <- function(p) ifelse(p < .001, "< .001", sub("^0", "", sprintf("%.3f", p)))
fmt_n <- function(x, d = 2) sub("^(-?)0\\.", "\\1.", sprintf(paste0("%.", d, "f"), x))

# Contrast weights keyed to an emmGrid's own row order
weights_for <- function(emm, pos, neg) {
  g <- as.data.frame(emm)
  lab <- paste(g$group, g$condition, sep = ".")
  w <- setNames(rep(0, length(lab)), lab)
  stopifnot(pos %in% lab, neg %in% lab)
  w[pos] <- 1
  w[neg] <- -1
  w
}
```

1. Load data and build composites

Each subscale is the mean of its two items, and each family is the mean of its subscales. With complete data this equals the item mean.

```r
raw <- read_excel(data_path)

needed <- c(unlist(binding_subscales), unlist(individualizing_subscales),
            ideology_col, group_col, condition_col)
miss <- setdiff(needed, names(raw))
if (length(miss)) stop("Missing columns: ", paste(miss, collapse = ", "))

sub_mean <- function(df, items) rowMeans(sapply(df[items], as.numeric), na.rm = TRUE)

# Map numeric codes or text labels onto canonical levels
to_level <- function(x, codes, text) {
  x   <- tolower(gsub("[^A-Za-z0-9]", "", as.character(x)))
  out <- names(codes)[match(x, as.character(codes))]
  alt <- names(text)[match(x, text)]
  factor(ifelse(is.na(out), alt, out), levels = names(codes))
}

grp_raw  <- raw[[group_col]]
cond_raw <- raw[[condition_col]]
data <- raw
data$group     <- to_level(grp_raw,  group_codes,     group_text)
data$condition <- to_level(cond_raw, condition_codes, condition_text)
data$ideology  <- as.numeric(raw[[ideology_col]])

unmapped <- is.na(data$group) | is.na(data$condition)
if (any(unmapped)) {
  cat("Rows dropped because group or condition could not be mapped:",
      sum(unmapped), "\n")
  print(table(group = grp_raw[unmapped], condition = cond_raw[unmapped], useNA = "ifany"))
}

data$binding <- rowMeans(sapply(binding_subscales, sub_mean, df = data), na.rm = TRUE)
data$individualizing <- rowMeans(sapply(individualizing_subscales, sub_mean, df = data),
                                 na.rm = TRUE)
```

```r
data <- data %>% filter(!is.na(group), !is.na(condition))
data$ideology_c <- data$ideology - mean(data$ideology, na.rm = TRUE)

if (id_col %in% names(data)) {
  cat("Duplicate IDs:", sum(duplicated(data[[id_col]])), "\n")
}
```

```
Duplicate IDs: 0
```

```r
kable(with(data, table(group, condition)), caption = "Cell sizes")
```

Table 1: Cell sizes

	Outgroup	Metaperception	Ingroup
Pro-life	89	85	89
Pro-choice	96	97	96

2. Descriptives (raw, unadjusted)

```r
desc <- data %>%
  group_by(group, condition) %>%
  summarise(n = n(),
            M_ind = mean(individualizing, na.rm = TRUE),
            SD_ind = sd(individualizing, na.rm = TRUE),
            M_bind = mean(binding, na.rm = TRUE),
            SD_bind = sd(binding, na.rm = TRUE),
            M_ideology = mean(ideology, na.rm = TRUE),
            .groups = "drop")
kable(desc, digits = 3)
```

group	condition	n	M_ind	SD_ind	M_bind	SD_bind	M_ideology
Pro-life	Outgroup	89	3.079	0.965	2.029	1.012	5.427
Pro-life	Metaperception	85	3.065	0.737	3.574	0.865	5.529
Pro-life	Ingroup	89	3.508	0.764	3.556	0.821	5.652
Pro-choice	Outgroup	96	2.120	0.788	3.335	0.940	2.677
Pro-choice	Metaperception	97	3.113	1.082	1.473	0.277	2.608
Pro-choice	Ingroup	96	3.841	0.811	2.099	0.827	2.719

The `M_ideology` column shows how far apart the groups sit on the covariate. An ideology-adjusted mean is evaluated at the grand mean (about the scale midpoint), a value few participants in either group actually hold.

3. Assumption checks

```r
# Homogeneity of regression slopes: does ideology relate to ratings
# differently across cells?
slopes_ind <- lm(individualizing ~ ideology_c * group * condition, data = data)
slopes_bin <- lm(binding ~ ideology_c * group * condition, data = data)
kable(Anova(slopes_ind, type = 2), digits = 3,
      caption = "Individualizing: covariate x factor interactions (Type II)")
```

Table 2: Individualizing: covariate x factor interactions (Type II)

	Sum Sq	Df	F value	Pr(>F)
ideology_c	2.719	1	3.687	0.055
group	0.483	1	0.654	0.419
condition	111.477	2	75.569	0.000
ideology_c:group	1.008	1	1.367	0.243
ideology_c:condition	3.959	2	2.684	0.069
group:condition	11.545	2	7.826	0.000
ideology_c:group:condition	6.752	2	4.577	0.011
Residuals	398.295	540	NA	NA

```
kable(Anova(slopes_bin, type = 2), digits = 3,
      caption = "Binding: covariate x factor interactions (Type II)")
```

Table 3: Binding: covariate x factor interactions (Type II)

	Sum Sq	Df	F value	Pr(>F)
ideology_c	0.063	1	0.094	0.760
group	26.229	1	39.055	0.000
condition	10.854	2	8.081	0.000
ideology_c:group	0.000	1	0.000	0.984
ideology_c:condition	5.923	2	4.410	0.013
group:condition	122.516	2	91.214	0.000
ideology_c:group:condition	0.095	2	0.071	0.932
Residuals	362.656	540	NA	NA

```
if (has_heplots) {
  print(heplots::boxM(cbind(individualizing, binding) ~ group * condition, data = data))
} else {
  cat("Install 'heplots' to run Box's M.\n")
}
```

```
 Box's M-test for Homogeneity of Covariance Matrices

data:  data
Chi-Sq (approx.) = 175.0249, df = 15, p-value = < 2.2e-16
```

Significant `ideology_c` interaction terms mean the single-slope ANCOVA assumption is violated. Section 8 checks whether that matters for the planned contrasts.

4. MANCOVA

Sum-to-zero contrasts and a centered covariate make every Type III test interpretable. The first table uses `car::Manova` (Type III). The second reproduces a sequential `manova()` call, whose interaction test is what the manuscript reports.

```
options(contrasts = c("contr.sum", "contr.poly"))

mlm_fit <- lm(cbind(individualizing, binding) ~ group * condition + ideology_c,
              data = data)
mancova_III <- Manova(mlm_fit, type = 3)
print(mancova_III, test = "Pillai")
```

```
Type III MANOVA Tests: Pillai test statistic
               Df test stat approx F num Df den Df    Pr(>F)
```

```
(Intercept)      1   0.95196   5389.9      2    544 < 2.2e-16 ***
group            1   0.08808     26.3      2    544 1.282e-11 ***
condition        2   0.22203     34.0      4   1090 < 2.2e-16 ***
ideology_c       1   0.00659      1.8      2    544    0.1653
group:condition  2   0.52545     97.1      4   1090 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
manova_seq <- manova(cbind(individualizing, binding) ~ group * condition + ideology,
                     data = data)
summary(manova_seq, test = "Pillai")
```

```
                Df  Pillai approx F num Df den Df Pr(>F)
group            1 0.17239   56.658      2    544 <2e-16 ***
condition        2 0.24342   37.762      4   1090 <2e-16 ***
ideology         1 0.00579    1.585      2    544 0.2059
group:condition  2 0.52545   97.104      4   1090 <2e-16 ***
Residuals      545
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5. Univariate ANCOVAs (corrected)

```
fit_ind <- lm(individualizing ~ group * condition + ideology_c, data = data)
fit_bin <- lm(binding ~ group * condition + ideology_c, data = data)

ancova_table <- function(fit, label) {
  a <- Anova(fit, type = 3)
  a <- a[rownames(a) != "(Intercept)", ]
  eta <- eta_squared(Anova(fit, type = 3), partial = TRUE, ci = .95)
  res_df <- a["Residuals", "Df"]
  a <- a[rownames(a) != "Residuals", ]
  data.frame(DV = label, effect = rownames(a),
             F = a$`F value`, df1 = a$Df, df2 = res_df,
             p = fmt_p(a$`Pr(>F)`),
             eta2_p = eta$Eta2_partial[match(rownames(a), eta$Parameter)],
             row.names = NULL)
}

ancova_corrected <- rbind(ancova_table(fit_ind, "Individualizing"),
                          ancova_table(fit_bin, "Binding"))
kable(ancova_corrected, digits = 3,
      caption = "Type III ANCOVA, sum contrasts, centered ideology")
```

Table 4: Type III ANCOVA, sum contrasts, centered ideology

DV	effect	F	df1	df2	p	eta2_p
Individualizing	group	0.182	1	545	.670	0.000
Individualizing	condition	70.139	2	545	< .001	0.205
Individualizing	ideology_c	3.613	1	545	.058	0.007
Individualizing	group:condition	28.640	2	545	< .001	0.095
Binding	group	51.622	1	545	< .001	0.087
Binding	condition	6.221	2	545	.002	0.022
Binding	ideology_c	0.093	1	545	.760	0.000
Binding	group:condition	221.930	2	545	< .001	0.449

For comparison, the original specification (treatment contrasts, uncentered ideology). Only the `group:condition` and `ideology` rows should match the corrected table; the `group` and `condition` rows were simple effects, not main effects.

```r
options(contrasts = c("contr.treatment", "contr.poly"))
fit_ind_old <- lm(individualizing ~ group * condition + ideology, data = data)
fit_bin_old <- lm(binding ~ group * condition + ideology, data = data)
old_tab <- rbind(
  data.frame(DV = "Individualizing", effect = rownames(Anova(fit_ind_old, type = 3)),
             F = Anova(fit_ind_old, type = 3)$`F value`),
  data.frame(DV = "Binding", effect = rownames(Anova(fit_bin_old, type = 3)),
             F = Anova(fit_bin_old, type = 3)$`F value`)
)
kable(subset(old_tab, !effect %in% c("(Intercept)", "Residuals")), digits = 2,
      row.names = FALSE, caption = "Original (treatment-contrast) Type III F values")
```

Table 5: Original (treatment-contrast) Type III F values

DV	effect	F
Individualizing	group	30.71
Individualizing	condition	7.15
Individualizing	ideology	3.61
Individualizing	group:condition	28.64
Binding	group	89.87
Binding	condition	102.25
Binding	ideology	0.09
Binding	group:condition	221.93

```r
options(contrasts = c("contr.sum", "contr.poly"))
```

6. Estimated marginal means

Adjusted to the grand-mean ideology.

```r
emm_ind <- emmeans(fit_ind, ~ group * condition)
emm_bin <- emmeans(fit_bin, ~ group * condition)
kable(as.data.frame(emm_ind), digits = 3, caption = "Individualizing EMMs")
```

Table 6: Individualizing EMMs

group	condition	emmean	SE	df	lower.CL	upper.CL
Pro-life	Outgroup	3.007	0.099	545	2.812	3.202
Pro-choice	Outgroup	2.189	0.096	545	2.001	2.377
Pro-life	Metaperception	2.988	0.102	545	2.787	3.189
Pro-choice	Metaperception	3.186	0.096	545	2.998	3.375
Pro-life	Ingroup	3.426	0.102	545	3.226	3.625
Pro-choice	Ingroup	3.909	0.095	545	3.721	4.096

```r
kable(as.data.frame(emm_bin), digits = 3, caption = "Binding EMMs")
```

Table 7: Binding EMMs

group	condition	emmean	SE	df	lower.CL	upper.CL
Pro-life	Outgroup	2.019	0.094	545	1.834	2.204
Pro-choice	Outgroup	3.345	0.091	545	3.167	3.524
Pro-life	Metaperception	3.562	0.097	545	3.371	3.753
Pro-choice	Metaperception	1.484	0.091	545	1.305	1.663
Pro-life	Ingroup	3.544	0.096	545	3.354	3.733

group	condition	emmean	SE	df	lower.CL	upper.CL
Pro-choice	Ingroup	2.109	0.090	545	1.932	2.287

7. Planned contrasts (preregistered)

Unadjusted p values. `d_gap` is the standardized difference between the two contrasts in a family (the manuscript's between-group gap in accuracy).

```
run_contrasts <- function(emm, fit, pairs, family) {
  wl <- lapply(pairs, function(pr) weights_for(emm, pr[1], pr[2]))
  ct <- contrast(emm, method = wl, adjust = "none")
  s  <- summary(ct, infer = c(TRUE, TRUE))
  sig <- sigma(fit)
  out <- data.frame(family = family, contrast = s$contrast,
                    M_diff = s$estimate, SE = s$SE,
                    CI_low = s$lower.CL, CI_high = s$upper.CL,
                    t = s$t.ratio, df = s$df, p = fmt_p(s$p.value),
                    d = s$estimate / sig)
  gap <- summary(eff_size(ct, sigma = sig, edf = df.residual(fit)))
  list(table = out, gap = data.frame(family = family,
                                     d_gap = gap$effect.size,
                                     CI_low = gap$lower.CL, CI_high = gap$upper.CL),
       ct = ct)
}

h1_pairs <- list(
  "Pro-choice ingroup vs Pro-life outgroup" = c("Pro-choice.Ingroup", "Pro-life.Outgroup"),
  "Pro-life ingroup vs Pro-choice outgroup" = c("Pro-life.Ingroup", "Pro-choice.Outgroup"))
h2_pairs <- list(
  "Pro-choice outgroup vs Pro-life metaperception" = c("Pro-choice.Outgroup", "Pro-life.Metaperception"),
  "Pro-life outgroup vs Pro-choice metaperception" = c("Pro-life.Outgroup", "Pro-choice.Metaperception"))

h1_ind <- run_contrasts(emm_ind, fit_ind, h1_pairs, "Individualizing")
h1_bin <- run_contrasts(emm_bin, fit_bin, h1_pairs, "Binding")
h2_ind <- run_contrasts(emm_ind, fit_ind, h2_pairs, "Individualizing")
h2_bin <- run_contrasts(emm_bin, fit_bin, h2_pairs, "Binding")

kable(rbind(h1_ind$table, h1_bin$table), digits = 3,
      caption = "H1: ingroup vs outgroup perception (opposing stance)")
```

Table 8: H1: ingroup vs outgroup perception (opposing stance)

family	contrast	M_diff	SE	CI_low	CI_high	t	df	p	d
Individualizing	Pro-choice ingroup vs Pro-life outgroup	0.901	0.147	0.612	1.190	6.129	545	< .001	1.039
Individualizing	Pro-life ingroup vs Pro-choice outgroup	1.236	0.151	0.940	1.532	8.202	545	< .001	1.425
Binding	Pro-choice ingroup vs Pro-life outgroup	0.091	0.139	-0.183	0.364	0.650	545	.516	0.110
Binding	Pro-life ingroup vs Pro-choice outgroup	0.198	0.143	-0.082	0.479	1.388	545	.166	0.241

```
kable(rbind(h1_ind$gap, h1_bin$gap), digits = 3, caption = "H1: between-group gap (d)")
```

Table 9: H1: between-group gap (d)

family	d_gap	CI_low	CI_high
Individualizing	-0.386	-0.923	0.150
Binding	-0.131	-0.667	0.405

```
kable(rbind(h2_ind$table, h2_bin$table), digits = 3,
      caption = "H2: outgroup perception vs metaperception (the accuracy tests)")
```

Table 10: H2: outgroup perception vs metaperception (the accuracy tests)

family	contrast	M_diff	SE	CI_low	CI_high	t	df	p	d
Individualizing	Pro-choice outgroup vs Pro-life metaperception	-0.799	0.150	-1.094	-0.504	-5.314	545	< .001	-0.921
Individualizing	Pro-life outgroup vs Pro-choice metaperception	-0.179	0.148	-0.470	0.112	-1.208	545	.228	-0.206
Binding	Pro-choice outgroup vs Pro-life metaperception	-0.217	0.143	-0.497	0.063	-1.520	545	.129	-0.263
Binding	Pro-life outgroup vs Pro-choice metaperception	0.535	0.141	0.258	0.811	3.803	545	< .001	0.650

```
kable(rbind(h2_ind$gap, h2_bin$gap), digits = 3, caption = "H2: between-group gap (d)")
```

Table 11: H2: between-group gap (d)

family	d_gap	CI_low	CI_high
Individualizing	-0.714	-1.253	-0.176
Binding	-0.913	-1.451	-0.376

Table 1 rows (Δ = metaperception − actual)

The emmeans contrast is *actual* − *metaperception*, so Δ flips its sign.

```
t1 <- rbind(h2_ind$table, h2_bin$table)
t1$judge <- ifelse(grepl("Pro-life metaperception", t1$contrast),
                   "Pro-life", "Pro-choice")
t1$Delta <- -t1$M_diff
t1$reading <- ifelse(t1$p != "< .001" & as.numeric(sub("< ", "", t1$p)) >= .05,
                     "Accurate", ifelse(t1$Delta > 0, "Overestimate", "Underestimate"))
kable(t1[, c("judge", "family", "Delta", "SE", "t", "df", "p", "reading")],
      digits = 2, row.names = FALSE)
```

judge	family	Delta	SE	t	df	p	reading
Pro-life	Individualizing	0.80	0.15	-5.31	545	< .001	Overestimate
Pro-choice	Individualizing	0.18	0.15	-1.21	545	.228	Accurate
Pro-life	Binding	0.22	0.14	-1.52	545	.129	Accurate
Pro-choice	Binding	-0.53	0.14	3.80	545	< .001	Underestimate

8. Robustness of the planned contrasts

Three specifications of the H2 accuracy contrasts:

- **A. Preregistered:** single ideology slope, means at grand-mean ideology (section 7).
- **B. No covariate:** raw cell means.
- **C. Separate slopes:** ideology allowed to relate differently to ratings in each condition, with each group evaluated at its own mean ideology.

If a cell's reading changes across specifications, report that in the SI.

```r
h2_under <- function(fit, emm, family, spec) {
  r <- run_contrasts(emm, fit, h2_pairs, family)$table
  r$spec <- spec
  r$Delta <- -r$M_diff
  r[, c("spec", "family", "contrast", "Delta", "SE", "t", "p")]
}

# B: no covariate
fit_ind_B <- lm(individualizing ~ group * condition, data = data)
fit_bin_B <- lm(binding ~ group * condition, data = data)

# C: separate slopes by condition, groups at their own mean ideology
fit_ind_C <- lm(individualizing ~ group * condition + ideology_c * condition, data = data)
fit_bin_C <- lm(binding ~ group * condition + ideology_c * condition, data = data)
emm_C <- function(fit) emmeans(fit, ~ group * condition,
                               cov.reduce = ideology_c ~ group)

rob <- rbind(
  h2_under(fit_ind, emm_ind, "Individualizing", "A. Preregistered"),
  h2_under(fit_ind_B, emmeans(fit_ind_B, ~ group * condition), "Individualizing", "B. No covariate"),
  h2_under(fit_ind_C, emm_C(fit_ind_C), "Individualizing", "C. Separate slopes"),
  h2_under(fit_bin, emm_bin, "Binding", "A. Preregistered"),
  h2_under(fit_bin_B, emmeans(fit_bin_B, ~ group * condition), "Binding", "B. No covariate"),
  h2_under(fit_bin_C, emm_C(fit_bin_C), "Binding", "C. Separate slopes")
)
rob <- rob[order(rob$family, rob$contrast, rob$spec), ]
kable(rob, digits = 3, row.names = FALSE,
      caption = "H2 accuracy contrasts across specifications (Delta = metaperception - actual)")
```

Table 12: H2 accuracy contrasts across specifications (Delta = metaperception - actual)

spec	family	contrast	Delta	SE	t	p
A. Preregistered	Binding	Pro-choice outgroup vs Pro-life metaperception	0.217	0.143	-1.520	.129
B. No covariate	Binding	Pro-choice outgroup vs Pro-life metaperception	0.239	0.122	-1.952	.051
C. Separate slopes	Binding	Pro-choice outgroup vs Pro-life metaperception	0.238	0.122	-1.957	.051
A. Preregistered	Binding	Pro-life outgroup vs Pro-choice metaperception	-0.535	0.141	3.803	< .001
B. No covariate	Binding	Pro-life outgroup vs Pro-choice metaperception	-0.557	0.121	4.614	< .001
C. Separate slopes	Binding	Pro-life outgroup vs Pro-choice metaperception	-0.549	0.120	4.568	< .001
A. Preregistered	Individualizing	Pro-choice outgroup vs Pro-life metaperception	0.799	0.150	-5.314	< .001
B. No covariate	Individualizing	Pro-choice outgroup vs Pro-life metaperception	0.945	0.130	-7.296	< .001
C. Separate slopes	Individualizing	Pro-choice outgroup vs Pro-life metaperception	0.946	0.129	-7.348	< .001
A. Preregistered	Individualizing	Pro-life outgroup vs Pro-choice metaperception	0.179	0.148	-1.208	.228
B. No covariate	Individualizing	Pro-life outgroup vs Pro-choice metaperception	0.035	0.128	-0.272	.786
C. Separate slopes	Individualizing	Pro-life outgroup vs Pro-choice metaperception	0.026	0.127	-0.207	.836

9. Reproduction checks against the manuscript

Each row compares a value computed above with the value printed in the manuscript.

```r
seq_stats  <- summary(manova_seq, test = "Pillai")$stats
pillai_int <- seq_stats["group:condition", "Pillai"]
ideo_F     <- seq_stats["ideology", "approx F"]
bind_int_F <- Anova(fit_bin, type = 3)["group:condition", "F value"]
h1 <- rbind(h1_ind$table, h1_bin$table)
h2 <- rbind(h2_ind$table, h2_bin$table)

checks <- data.frame(
  quantity = c("Pillai, group x condition",
               "Ideology covariate, multivariate F",
               "Binding univariate interaction F",
               "Indiv: pro-choice ingroup - pro-life outgroup",
               "Indiv: pro-life ingroup - pro-choice outgroup",
               "Indiv: pro-choice outgroup - pro-life meta",
```

```
               "Indiv: pro-life outgroup - pro-choice meta",
               "Bind: pro-choice outgroup - pro-life meta",
               "Bind: pro-life outgroup - pro-choice meta"),
  manuscript = c(.525, 1.59, 221.93, 0.90, 1.24, -0.80, -0.18, -0.22, 0.53),
  computed = c(pillai_int, ideo_F, bind_int_F,
               h1$M_diff[h1$family == "Individualizing"], h2$M_diff)
)
checks$match <- abs(round(checks$computed, 2) - checks$manuscript) < .006
kable(checks, digits = 3)
```

quantity	manuscript	computed	match
Pillai, group x condition	0.525	0.525	TRUE
Ideology covariate, multivariate F	1.590	1.585	FALSE
Binding univariate interaction F	221.930	221.930	TRUE
Indiv: pro-choice ingroup - pro-life outgroup	0.900	0.901	TRUE
Indiv: pro-life ingroup - pro-choice outgroup	1.240	1.236	TRUE
Indiv: pro-choice outgroup - pro-life meta	-0.800	-0.799	TRUE
Indiv: pro-life outgroup - pro-choice meta	-0.180	-0.179	TRUE
Bind: pro-choice outgroup - pro-life meta	-0.220	-0.217	TRUE
Bind: pro-life outgroup - pro-choice meta	0.530	0.535	TRUE

If the two ingroup-vs-outgroup rows fail with the values swapped, the stance codes in the configuration are reversed.

10. Figure 2

Ideology-adjusted scores (unit-preserving adjustment to the covariate mean), with model-based EMMs ± 95% CI. `coord_cartesian` zooms without discarding observations, so violins use all the data.

```
b_ind <- coef(fit_ind)[["ideology_c"]]
b_bin <- coef(fit_bin)[["ideology_c"]]

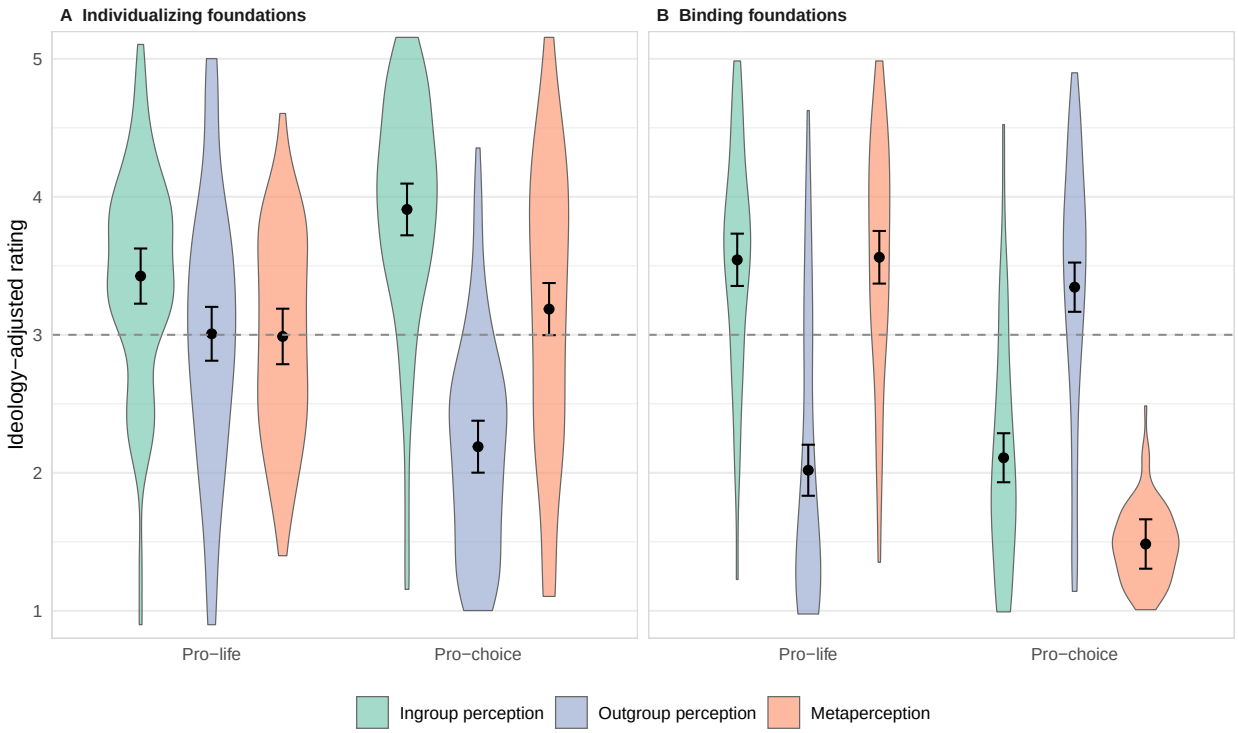
plot_df <- data %>%
  mutate(Individualizing = individualizing - b_ind * ideology_c,
         Binding = binding - b_bin * ideology_c) %>%
  select(group, condition, Individualizing, Binding) %>%
  pivot_longer(c(Individualizing, Binding), names_to = "family", values_to = "value")

emm_df <- rbind(
  cbind(family = "Individualizing", as.data.frame(emm_ind)),
  cbind(family = "Binding", as.data.frame(emm_bin))
)

cond_levels <- c("Ingroup", "Outgroup", "Metaperception")
cond_labels <- c("Ingroup perception", "Outgroup perception", "Metaperception")
plot_df$condition <- factor(plot_df$condition, cond_levels, cond_labels)
emm_df$condition  <- factor(emm_df$condition,  cond_levels, cond_labels)
fam_levels <- c("Individualizing", "Binding")
plot_df$family <- factor(plot_df$family, fam_levels,
                         c("A  Individualizing foundations", "B  Binding foundations"))
emm_df$family  <- factor(emm_df$family, fam_levels,
                         c("A  Individualizing foundations", "B  Binding foundations"))

pd <- position_dodge(width = 0.8)
ggplot(plot_df, aes(group, value, fill = condition)) +
  geom_violin(position = pd, width = 0.75, alpha = 0.6, colour = "grey40",
              linewidth = 0.3) +
  geom_errorbar(data = emm_df, aes(x = group, ymin = lower.CL, ymax = upper.CL,
                                   group = condition),
                inherit.aes = FALSE, position = pd, width = 0.15) +
  geom_point(data = emm_df, aes(x = group, y = emmean, group = condition),
```

```
            inherit.aes = FALSE, position = pd, size = 2) +
  geom_hline(yintercept = 3, linetype = "dashed", colour = "grey55") +
  facet_wrap(~ family) +
  scale_fill_manual(values = c("#66c2a5", "#8da0cb", "#fc8d62"), name = NULL) +
  coord_cartesian(ylim = c(1, 5)) +
  labs(x = NULL, y = "Ideology-adjusted rating") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold", hjust = 0),
        panel.grid.major.x = element_blank(),
        panel.border = element_rect(fill = NA, colour = "grey85"))
```



11. Manuscript-ready text

```
pil <- summary(manova_seq, test = "Pillai")$stats["group:condition", ]
cat("**Interaction:** `Pillai's Trace $=", fmt_n(pil["Pillai"], 3), "$, $F(",
    pil["num Df"], ", ", pil["den Df"], ") = ", sprintf("%.2f", pil["approx F"]),
    "$, $p ", ifelse(pil["Pr(>F)"] < .001, "< .001", paste("=", fmt_n(pil["Pr(>F)"], 3))),
    "$`\n\n", sep = "")
```

Interaction: `Pillai's Trace $=.525$, $F(4, 1090) = 97.10$, $p < .001$`

```
cat("**Table 1 (Study 2) rows:**\n\n")
```

Table 1 (Study 2) rows:

```
for (i in seq_len(nrow(t1))) {
  r <- t1[i, ]
  tval <- if (r$reading == "Accurate") "---" else sprintf("%.2f (%d)", abs(r$t), r$df)
  cat(sprintf("- `%s & %s & $%s%.2f$ & %s & %s & %s \\\\`\n",
```

```
            r$judge, r$family, ifelse(r$Delta >= 0, "+", "-"), abs(r$Delta),
            tval, sub("< ", "$<$ ", r$p), r$reading))
}
```

- Pro-life & Individualizing & $+0.80$ & 5.31 (545) & $<$ .001 & Overestimate \\
- Pro-choice & Individualizing & $+0.18$ & --- & .228 & Accurate \\
- Pro-life & Binding & $+0.22$ & --- & .129 & Accurate \\
- Pro-choice & Binding & -0.53 & 3.80 (545) & $<$ .001 & Underestimate \\

Session info

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] knitr_1.51       effectsize_1.0.2 emmeans_2.0.3    car_3.1-5
[5] carData_3.0-6    ggplot2_4.0.3    tidyr_1.3.2      dplyr_1.2.1
[9] readxl_1.5.0.1

loaded via a namespace (and not attached):
 [1] generics_0.1.4     rgl_1.3.36         digest_0.6.39      magrittr_2.0.5
 [5] evaluate_1.0.5     grid_4.6.0         estimability_2.0.0 RColorBrewer_1.1-3
 [9] mvtnorm_1.4-1      fastmap_1.2.0      cellranger_1.1.0   jsonlite_2.0.0
[13] backports_1.5.1    Formula_1.2-5      purrr_1.2.2        scales_1.4.0
[17] abind_1.4-8        cli_3.6.6          rlang_1.2.0        base64enc_0.1-6
[21] withr_3.0.3        yaml_2.3.12        otel_0.2.0         tools_4.6.0
[25] datawizard_1.3.1   heplots_1.8.1      bayestestR_0.18.1  broom_1.0.13
[29] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    htmlwidgets_1.6.4
[33] MASS_7.3-65        insight_1.5.2      pkgconfig_2.0.3    pillar_1.11.1
[37] gtable_0.3.6       glue_1.8.1         xfun_0.59          tibble_3.3.1
[41] tidyselect_1.2.1   rstudioapi_0.19.0  parameters_0.29.2  farver_2.1.2
[45] htmltools_0.5.9    labeling_0.4.3     rmarkdown_2.31     compiler_4.6.0
[49] S7_0.2.2
```